

Virtual Reality-Based Retrospective Think Aloud (VR-RTA): A Novel Method for Studying Offender Decision-Making

Carmen-Silva Sergiou¹ Dominik Gerstner² Claire Nee³ Henk Elffers⁴
Jean-Louis van Gelder²

¹Leiden University, ²Max Planck Institute for Crime, Security and Law, ³University of Portsmouth,
⁴NSCR

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ABSTRACT

This article describes a novel multimodal method to examine decision-making: Virtual Reality-Based Retrospective Think Aloud (VR-RTA). This method taps directly into the offenders' perspective and aims to enhance memory recall. We apply this method among a sample of incarcerated burglars (N=200) who scouted neighborhoods in immersive virtual reality to explore possibilities for committing a burglary. Subsequently, they viewed back a screen recording of their scouting process and simultaneously 'thought aloud' about their assessment of the environment and the decision-making strategies they employed. Emerging themes were then further examined in an interview. Rich and detailed insights into participants' interpretation of the environment and their decision-making strategies were obtained. VR-RTA assisted in verbalizing automated cognitive processes and increased engagement by building rapport. That is, the method allows for capturing in the moment considerations underlying decision making during the crime commission process. We conclude that the VR-RTA method is a useful complement to other methods of data collection, such as interviews and survey approaches, and also enhance virtual reality-based research designs.

Introduction

Those with prolific experience with offending are uniquely placed to provide insight into the development of that behavior (Topalli et al. 2020). In criminology, interviews have long been regarded as the go-to method to tap into this insight (e.g., Maruna, 2001; Brookman et al. 2011; Miller et al. 2013; Sampson and Laub, 2003; Wright and Decker, 1994, Wright et al. 1995). However, interviews are subject to the inevitable degradation of memory recall. As time between the commission of a crime and a subsequent interview passes, memory becomes blurred, details are lost, and timing is warped, diminishing the accuracy and completeness of the information obtained (Nee 2010; van Gelder 2023). To address these challenges, researchers would ideally be present at the moment of offending or right after the event, but ethical, practical, and safety considerations generally render this option unfeasible (van Gelder 2023).

In this article, we describe and test a novel method, Virtual Reality-Based Retrospective Think-Aloud (VR-RTA), that integrates VR technology with so-called think-aloud protocols to overcome some of the limitations plaguing conventional research approaches, such as interviews and surveys. Think-aloud protocols involve research participants articulating their thoughts while carrying out a specific task or immediately following it (Simon & Ericsson, 1984). This method alleviates some of the constraints affecting interviews and other retrospective methods and allows researchers to gain more direct insight into offender decision-making. In the present study, we tested the VR-RTA method among a sample of incarcerated burglars who were invited to scout different neighborhoods for opportunities for burglary. This process was screen-recorded and played back to them immediately following the scouting task. Throughout the play back, the burglars were asked to "think aloud" about their decision-making during the scouting process.

Below we first describe the cognitive underpinnings of think-aloud protocols and the use of VR technology in crime research, prior to presenting our study and discussing findings. We conclude with a discussion of the more general potential of the VR-RTA method for crime research.

Think-Aloud Protocol Versus Related Methods

In research applying think-aloud protocols (TAPs), participants are instructed to think out loud whilst completing a task, and to verbalize everything that goes through their minds while doing so (Ericsson and Simon 1993; Güss 2018). When individuals are asked to report on their decision-making during an action out of context, they commonly tend to rely on implicit theories about how the world works in general, which may not accurately reflect how their actual cognitions drive their behavior (Branch 2000; Eccles and Aarsal 2017). By verbalizing thoughts during task performance, this limitation is addressed and real-time insight into the reasoning process is gained (Leighton 2017). Furthermore, people may provide inaccurate reports about cognitions when asked about them retrospectively because they have only limited conscious access to them (Nisbett and Wilson 1977; Wolcott and Lobczowski 2021). By analyzing thought processes that occur *during* or *right after* an activity, TAPs can reveal information about the decisions people make, how they interact with their environment, and what factors influence how they reach their goal (Wolcott and Lobczowski 2021). Lastly, as highlighted by Nisbett and Wilson (1977), people often find it challenging to explain how and why they perceive or remember things. However, people are generally able to describe their own actions, as occurs during a TAP. Overall, TAPs allow for a more effortless and detailed (re)construction of actions, and can provide a picture of relevant cognitive processes in real time (Reinhart et al. 2022).

TAPs exist in two different modalities; concurrent and retrospective. In concurrent think-aloud approaches, people think out loud whilst performing a task, and this is followed by a post-task interview (e.g., Meenaghan et al. 2018). The retrospective think-aloud (RTA) approach, which is the focus of this study, involves people first completing a task and subsequently reflecting on it while a recording of their performance is played back to them (Ericsson and Simon 1993; Fox et al., 2011; Hertzum et al., 2009). An advantage of retrospective over concurrent TAPs is that the former carry less cognitive load and do not interfere with task performance. Consequently, they allow for the preservation of natural behavior (van den Haak et al., 2004).

Although a common method for studying a range of different behaviors (Gregg et al., 2017; Kesler et al., 2016; Malek et al., 2017), the use of TAPs, both concurrent and retrospective, in crime research has thus far been limited. Some studies have used revisiting recent crime scenes as a method of enhancing participant engagement and increasing ecological validity (Cromwell et al. 1991; Wright and Decker 1994). However, without the added bonus of the re-enactment of the behavior under study, these methods still remain subject to the well-documented limitations of human memory (Nee et al. 2019). The advent of VR technology has provided novel possibilities to overcome these limitations.

Moving closer to the action: The use of VR in crime research

Immersive VR allows for the creation of realistic simulations of criminogenic environments, which offenders can navigate to demonstrate decision-making under circumstances resembling those of a controlled behavioral experiment (van Gelder 2023; van Gelder et al., 2014). Several VR studies on criminal decision-making have emerged in recent years, mostly focusing on burglars (e.g., Meenaghan et al. 2018; Nee et al. 2019; van Sintemaartensdijk 2022). In one of these studies, Meenaghan et al. (2018) pioneered the use of concurrent TAPs to study incarcerated burglars' experiences while undertaking a simulated burglary on a laptop computer. This non-immersive study underscored the added value of TAPs and demonstrated how this method can help elicit detailed information on burglar decision-making.

Virtual Reality-Based Retrospective Think-Aloud (VR-RTA), which was developed and tested in the present study, is a technique that builds on the work of Meenaghan and colleagues (2018). The retrospective nature of the VR-RTA method reduces cognitive load and minimizes reactivity compared to concurrent TAPs. That is, the verbal explanations offenders provide when viewing back the virtual re-enactment of their actions, allow for a discussion of motivations and other factors influencing burglar behavior in their real-life experiences, without disrupting the actual process.

Beyond the creation of a realistic and immersive virtual version of a real world criminogenic environment, VR-RTA also enables leveraging the benefits of a first-person perspective. Previous studies have demonstrated that, compared to a third-person perspective, taking a first-person view enhances both detail and accessibility in memory recall (Bergouignan et al., 2014; Glenberg & Hayes, 2016; Repetto et al., 2016). Having people perform a certain behavior and subsequently observe that behavior from a first-person perspective stimulates those parts of the brain that trigger action memory—i.e., the way human memories are constructed is shaped by the perceptual and motor brain systems (Barsalou 2008; Glenberg 1997; Glenberg and Hayes 2016; Shapiro 2011).

In short, re-enacting facilitates the mental recall of past incidents (Meade et al. 2019), and re-enactment from a first-person perspective connects such experiences more easily to other memories and sensory information (Glenberg and Hayes 2016). In the present study, by scouting virtual residential neighborhoods and engaging in the active navigation of these neighborhoods, participants re-enacted prior behavior, thereby aiding the retrieval of memories related to their past burglaries in the real world.

The Current Study

In this study, we invited a sample of incarcerated burglars to navigate two different virtual neighborhoods, with the task of looking for opportunities for a burglary. One of the virtual neighborhoods was an exact virtual copy of an existing neighborhood in a city in the Netherlands, the other was designed to resemble a typical middle-class Dutch neighborhood. For each participant, the scouting process in one of the two neighborhoods was screen-recorded and subsequently played back to them. During the replay, they were instructed to think aloud

about the scouting process and to elaborate on the decisions they made throughout this process. The objective was to stimulate disclosure of information and to gain more insight into their thinking while “on the job.” Themes that emerged during the think-aloud process were further examined in a subsequent semi-structured interview.

To this end, the VR system was equipped with an integrated eye-tracking system, which recorded what features of the environment the burglars were focusing on and for how long. This allowed for objectively establishing the extent to which specific features, such as deterrent or attractive cues, were noticed and paid attention to by the burglars. We examine to what extent burglars in our study mentioned such cues in the VR-RTA and interview. Overall, research was conducted to explore the merits of the VR-RTA method by addressing two research questions: 1) To what extent do the present results align with established findings from prior research? and 2) To what extent does the VR-RTA complement and broaden the information extracted from the interview and eye-tracking data?

Method

Participants

Participants were 200 incarcerated male burglars (age range 19-61, $M_{age} = 33$) with different degrees of burglary experience serving prison sentences for various offences (not only burglary) in the Netherlands. Participants could be included if they were 18 years or older, had committed at least 5 burglaries in their lifetime, did not have an epileptic history, or were currently using heavy medication (such as anti-psychotics). Involvement in burglary was checked through previous conviction data, prison staff, and self-reported offending history. Participants were recruited in four different male-only prisons. Recruitment happened through leafletting prison wings, and through referral by other inmates and prison staff. In exchange for their cooperation, participants were paid €5 in line with prison policy. Ethical approval was obtained from the Ethics Committee of the University of Leiden (ECPW-2022/372). All participants signed an informed consent form.

Procedure

Data collection occurred in a dedicated room in the participating prison with only the researcher and the participant present. Participants received detailed information on confidentiality and were given the opportunity to ask questions prior to participating. After giving consent, each participant scouted two different virtual neighborhoods. For both neighborhoods they were instructed to scout it as if they intended to commit a burglary there. After scouting the second neighborhood, participants filled out a survey with questions regarding feelings of presence, deterrence, cybersickness, and other variables relevant to the larger Virtual Burglary Project (results reported elsewhere).¹ Following this, the think aloud-task was conducted. The session ended with a semi-structured interview and a brief survey with questions about burglary expertise. Subject to participant consent (90%), verbalizations during the think-aloud process and interview were audio-recorded

(and deleted after transcription). Following the interview and survey, participants were debriefed and support was offered for any issues raised. The entire session took 45-60 minutes.

VR-RTA instructions

The virtual scouting process in one of the two neighborhoods was screen-recorded based on random selection (1:1).² The recording was played back to participants after finishing scouting both neighborhoods. Before starting the playback, participants received the following instructions from the experimenter:

“I would like to learn a bit more about how you evaluate a neighborhood. I am particularly interested in what things you find important when you make decisions related to burglary. We will use a method that is called the ‘think-aloud’ method. I will now show you the recording of your VR experience in the first [second] neighborhood. The idea is that you simply tell me whatever comes to mind when you watch the recording. Just say everything that comes to mind, irrespective of whether you think it is relevant or not. I may also ask you some questions during the replay of the recording.”

The experimenter aimed to minimize interference during the process. Only in cases where participants did not verbalize any thoughts, prompts were used to stimulate disclosure. Examples of prompts included questions about the participant's chosen direction, actions taken at specific locations, prolonged periods of non-movement, and reasons for changing direction.

Materials

Virtual Environment

The virtual neighborhoods A and B (see Figures 1 and 2) were developed using the Unity Pro engine (version 2017.3.1f1) and viewed through the VIVE Pro Eye head-mounted display, which is equipped with built-in eye tracking technology. Participants wore headphones delivering immersive audio and navigated the virtual environment using a game controller.



Figure 1. Top view, aerial view, and first person view of Neighborhood A in the virtual environment.



Figure 2. Top view, aerial view, and first person view of Neighborhood B in the virtual environment.

Eye tracking

In both neighborhoods, a set of items (henceforth ‘Easter-Eggs’), that may act as burglary attractors (e.g., open window, ladder, packaging of expensive items) or deterrents (e.g., beware of dog sign, alarm box) were distributed across the neighborhoods (see Figure 3). We used eye tracking to assess whether these Easter-Eggs had been noticed by participants. Eye tracking algorithms use specific criteria to determine where a person is looking to help understand what draws a person's attention and what information they are actively processing in their mind.



Figure 3. Easter-Eggs in neighborhood A. Beware of the Dog'-sign, ladder below front window and the placement of the Easter-Eggs.

Spatial patterns

X and y coordinates of the participants in the neighborhoods were tracked per second. These data yielded time spent (in seconds) in the virtual neighborhood, walking patterns, as well as the distance travelled (in meters). Spatial data allows for generating heat maps that reflect how participants move through a neighborhood and provide insight into scouting strategies.

Survey

The scouting process for each of the neighborhoods was followed by a brief survey. Two items were used to assess the likelihood of committing a burglary: "I would break into this neighborhood" and "This neighborhood is not attractive for burglary." Five items assessed perceived guardianship, for example, "Residents in this neighborhood are vigilant" and "I had the feeling of being watched when I walked around." The items were answered on a Likert-type scale ranging from 1 (Strongly disagree) to 5 (Totally agree). Each

of the scales showed acceptable alpha reliability (Alpha, Risk = 0.77 / Spearman-Brown, Intention = 0.64 / Alpha, Guardianship = 0.65).

Semi-structured interview

Three items in the semi-structured interview that followed the VR-RTA asked about participants' approaches to committing burglaries in real-life in the period(s) when they were active as burglars prior to incarceration. The interview questions were: "How do you decide on where you burgle?", "What things about a house or in a neighborhood attract you?", "What things about a house or in a neighborhood deter you?".

Coding strategy of the RTA-data

RTA data were analyzed following a thematic approach (Braun and Clarke 2006), using MAXQDA 2022 software (VERBI Software 2021). Coding was performed by two independent coders. Inter-coder agreement was 95%, which can be considered very high (Hallgren 2012). Disagreements between coders were resolved through establishing consensus.

During the coding process, burglary cues were classified into codes. In turn, cues reflected aspects that burglars talked about during the RTA task. The analysis consisted of six phases (see Figure 4). Phase 1 involved familiarization with the RTA data and understanding of possible patterns emerging from these data. Phase 2 involved the generation of initial codes (features of the data that seemed of interest). "Codes identify a feature of the data [...] that appears interesting to the analyst" and can be assessed to gain a better understanding of the phenomena under study (Braun and Clarke 2006, p. 18; Michel-Villarreal et al. 2021). The outcome of this phase consisted of a list of independent codes across the data. Phase 3 entailed reviewing these codes and grouping them into "themes." In this phase, consideration was given to "how different codes may combine to form an overarching theme" (Braun and Clarke 2006:18). Such themes can be described as significant concepts that provide a link to attach substantial portions of data together (DeSantis and Ugarriza 2000). Subsequently, phase 4 consisted of reviewing the "coded data extracts for each theme to consider whether they appear to form a coherent pattern" (Braun and Clake 2006:9). Phase 5 involved naming and defining the themes to group the coded data, and phase 6 comprised the writing up of the categories and subcategories of burglary cues (see Figures 5 and 6).

The initial aim of analyzing the RTA data was to investigate whether burglary cues identified in prior research (Nee 2015; Peeters 2013) also emerged in the current study. Additionally, RTA data was utilized to highlight the usefulness of retrospective thinking exercises in supplementing interviews, focusing on burglars' decision-making processes and the characteristics that make houses appealing targets.

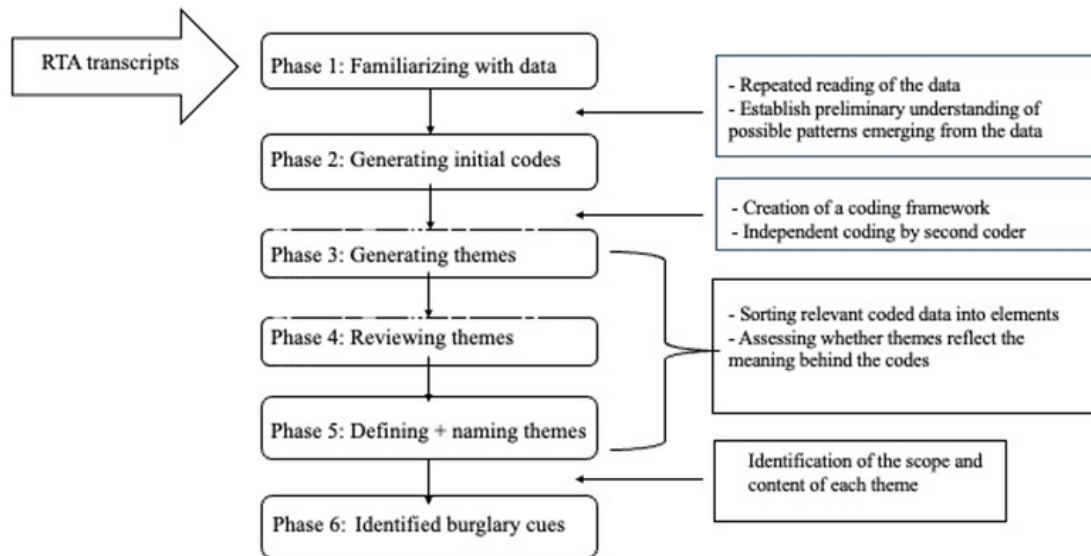


Figure 4. Thematic analysis process (adapted from Michel-Villarreal et al., 2021).

Results

The results are organized into two different parts that each address a research question. In the first part, we examine to what extent our findings regarding burglary cues align with previous research. The second part explores to what extent information extracted with the VR-RTA method complements the interview and eye-tracking data.

Research Question 1: Do RTA findings regarding burglary cues align with prior research?

Prior research has identified four different categories of cues related to burglar target selection: layout, security, occupancy, and affluence (for reviews see Nee 2015; Peeters 2013). Layout cues regard features such as degree of cover, presence of escape routes, and surveillability. Security cues include features related to security measures and target hardening, such as the presence of alarms and cameras, the quality of doors, and window locks. Occupancy cues are features signaling the presence of people in the immediate environment, such as the presence of a car, or light inside a house. Affluence cues include the availability of expensive items, the size of a house, and decor. We examined whether and how frequently these cue categories were mentioned during the RTA task. Figure 5 provides an overview of the number of mentions of the four different burglary categories, and shows how many of the participants mentioned them. In addition, Figure 6 displays the categories and their subcategories, and the percentage of participants that mentioned them.

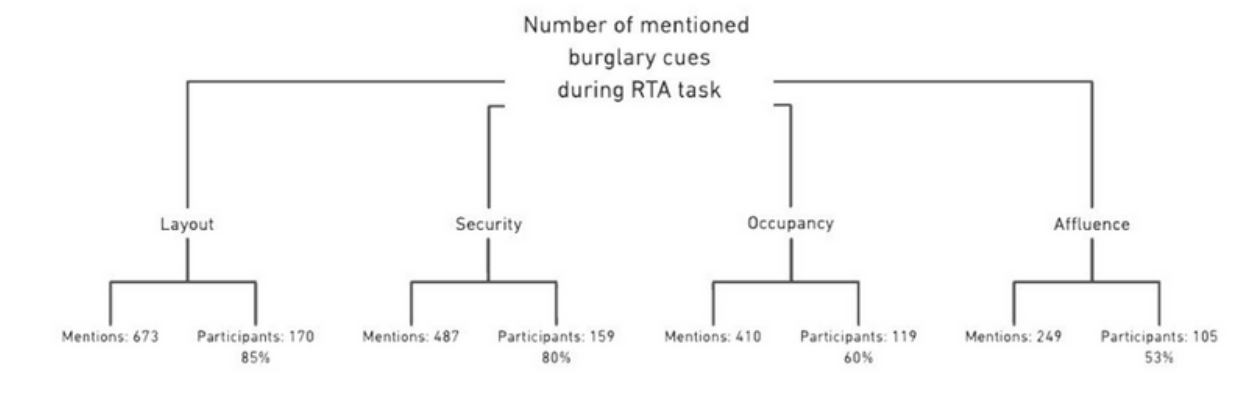


Figure 5. Number of mentions per participant (%) and per burglary category, total of 1,991 mentions across participants ($N=200$).

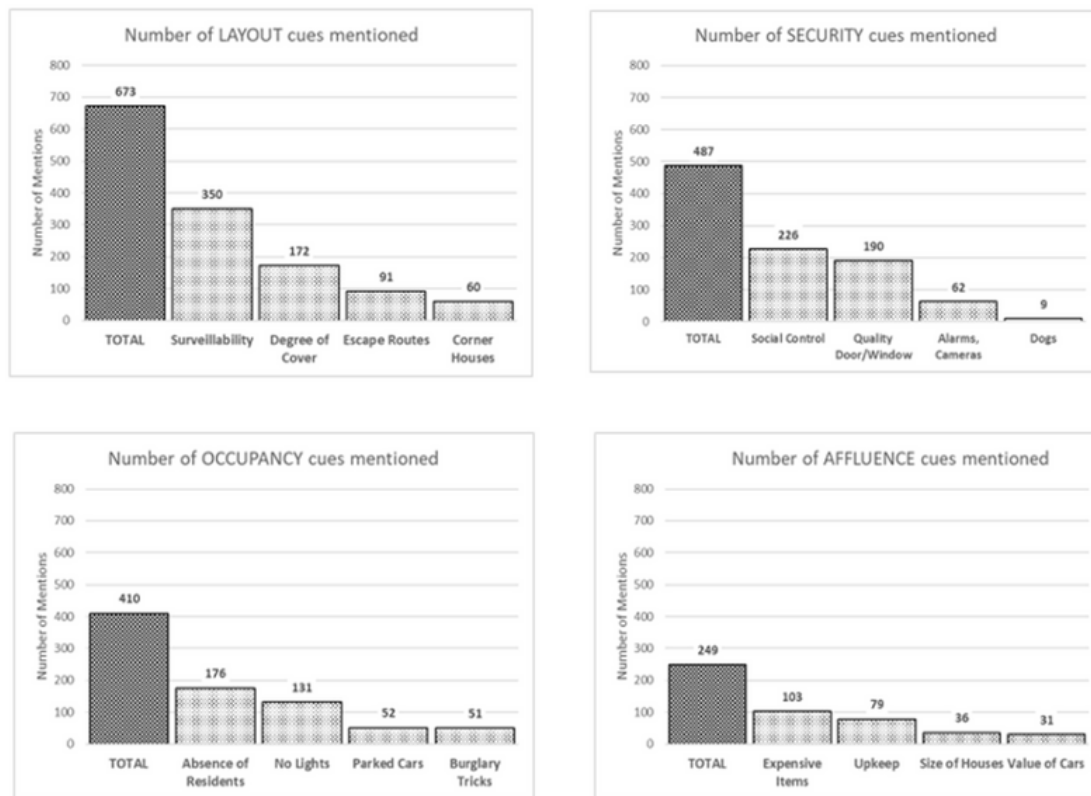


Figure 6. Number of mentioned burglary cues of each subcategory across participants ($N=200$).

Layout

In previous studies (e.g., Langton and Steenbeek 2017; Nee and Meenaghan 2006), layout cues were found to significantly influence burglars' decision-making. In the current study, layout cues were mentioned most often

(170 participants, 85%; see Figure 5), with cues mentioned predominantly relating to the subcategory *surveillability* (see Figure 6). Burglars mentioned lines of sight and the ease of being spotted in particular (e.g., by neighbors or passers-by). In addition, and related to surveillability, *degree of cover* and *escape routes* were also frequently mentioned, both often as a first remark upon entering the neighborhood. Quotes are provided in Table 1.

Table 1. Examples of layout cues mentioned

Escape routes

"You should also carefully consider the side streets in a neighborhood; is it located on a busy road or not, how easy is it to escape? You need to determine your escape routes for the car." (Participant 121)

Surveillance

"I noticed this was a difficult neighborhood. There is light everywhere and people are watching you from everywhere. You get that in cities. I used to go to small villages, where houses and farms lie 500 meters apart from each other, so nobody can see you doing things." (Participant 20)

Degree of Cover

"You just stand out very quickly here, when you walk down that street... the other neighborhood was much darker, there you have more alleyways. More places to hide." (Participant 168)

Corner houses

"Corner houses, actually always, there you only have one neighbor. Furthermore, you also know that corner houses are more expensive." (Participant 68)

Security

Security cues were also mentioned frequently (159 participants, 80% of the sample). Predominant subcategories mentioned related to *Quality of doors/windows* (e.g., estimated time required to break open locks, target hardening). These findings are congruent with results reported in the research literature on security cues (e.g., Newton et al., 2008). Furthermore, burglars paid particular attention to *social control* (i.e., the presence of people in the neighborhood who may act as guardians). They talked about social control systems

that might be active, such as neighborhood watches, vigilant parents, and communication between neighbors. Other security measures, such as cameras, alarms, or the possible presence of a dog were also mentioned, although less frequently.

Table 2. Examples of security cues mentioned

Quality of doors and window locks

“In principle this neighborhood is quite well secured; high doors, fences. You are always looking at which type of locks people have, which type of door. Is it made of wood or plastic? Is there a window open, or casement that is easy access with a screwdriver? These days a lot of houses have plastic window frames, (...) very easy to get in, same with doors. Materials matter a lot.” (Participant 67)

Alarms/Cameras

“People are guarding their homes with a smart doorbell nowadays, and then they will get instant messages on their phone. Those doorbells can also record sound from far away, so you need to always stop talking when you are with others, because then they will have evidence.” (Participant 22)

Dogs

(..) “This house has a clearly visible sign that says, ‘Beware of the dog’. This makes people that notice this sign automatically more cautious.” (Participant 41)

Social Control

“(...) because you can just see from this residential area that many people can look out of their windows; those kinds of people live here. People here look out for each other, pay attention to things that stand out.” (Participant 53)

Occupancy

The (possible) presence of residents was mentioned third most (119 participants, 60%), aligning with ample prior evidence indicating that burglars prefer unoccupied houses over occupied ones (Coupe and Blake 2006; Hearndon and Magill 2004; Wright and Decker 1994). Frequently mentioned subcategories to establish occupancy were lights being on in a house or cars being parked in front of it. Since participants were told prior to entering the VR that it was 5:30 PM in the virtual neighborhood, they indicated that this was a time that many residents would be home or about to come home, making this a sub-optimal time for committing

burglary. Additionally, burglars mentioned using tricks or technology to monitor residents' routines, for example by sticking small items (e.g., toothpicks) in the door, using a hidden camera to register residents' routines, or ringing the doorbell with a pretext.

Table 3. Examples of occupancy cues mentioned

Absent Residents

"I was looking whether I would see indications that someone is at home or not. You can determine that by looking at the lights or if there are parked cars or not. If the lights are off, you assume residents are not at home, but you always check if there is also movement. Lights on are not a direct indication that someone is home, but if the car is there, that is a good indicator, so you always check that first." (Participant 26)

Burglary tricks

"I would just ring the bell to see if the residents were home. If they would open, I would just ask 'does Johnny live here?' Easy. Or just stick something in the door like a toothpick." (Participant 80)

No lights

"Look, this house has all the lights off in the back and in the front, so no one is home, at least that's what you assume." (Participant 165)

Parked cars

"Well, those parking spaces actually indicated that no one was home or visiting." (Participant 1)

Affluence

Of the four categories, affluence cues were mentioned least often (105 participants, 53%). In this category, burglars primarily focused on valuable items that were visible from outside a house to determine wealth. Upkeep was also frequently mentioned within this category. House size and the value of parked cars were also remarked on, but less frequently. These findings align with previous research by Peeters (2013), who reported mixed results on how affluence influences burglar risk-taking and target selection.

Table 4. Examples of affluence cues mentioned

Expensive items

“I saw some iMac boxes laying outside the house. Then you know they are spending money. Who has and who hasn’t, that is the first thing you need to find out” (Participant 119)

Upkeep Décor

“Green Egg (BBQ) in the garden, some have a jacuzzi in their backyard, expensive garden furniture. So, these people have money.” (Participant 153)

Size Houses

“No man, I wouldn’t break into this neighborhood (...). It’s a working-class neighborhood. If you’re going for burglary, you’re going for money, and that’s not found here.” (Participant 197)

Value Cars

(..) “You also look at the cars, what value they have. Here, I see a beautiful Volvo, and here, a nice Mercedes.” (Participant 64)

Combined, the four different categories of burglary cues (i.e., layout, security, occupancy, affluence) that had been identified in earlier research, also emerged in the present study. In line with prior work, layout cues were most frequently mentioned with an emphasis on surveillance. Other frequently mentioned subcategories, such as social control, target hardening, and presence of residents, also aligned with previous work (e.g., Nee et al. 2019; Peeters 2013). Research question 2: VR-RTA as a method to complement and extend information from the interview and VR data

We have thus far focused on validating the VR-RTA method by relating our findings to findings from the broader documented burglary cue literature. We now turn to demonstrating how VR-RTA can complement eye tracking data and spatial patterns. Furthermore, we compare VR-RTA data with responses to interview questions to explore how the former can be used as a tool for eliciting information.

Research Question 2: Can VR-RTA Complement Interview and VR Eye-tracking Data?

In this section we will focus on exploring whether VR-RTA can usefully complement eye-tracking data and spatial patterns. This includes a comparison of VR-RTA data with responses to interview questions to explore how the former can be used as a tool for eliciting information.

Combining eye tracking with RTA

Eye-tracking data provide information about what aspects of a setting or environment people pay attention to. In this sense, eye-tracking technology allows researchers to see through the eyes of burglars in an almost literal way. Here eye-tracking was used to identify which Easter Eggs (i.e., items that may act as burglary attractors or deterrents (see Figure 3), that were dispersed throughout both neighborhoods, were noticed by participants. Eye-tracking data were linked to VR-RTA data to deepen our understanding of the gaze patterns (i.e., capturing more of the *why* underlying eye-tracking data).

Correlation analyses revealed significant relationships between for the amount of Easter Eggs spotted and the intention to burgle (Neighborhood A: $r = .25$, $p = .00$; Neighborhood B: $r = .16$, $p = .02$), and between spotted Easter Eggs and perceived feelings of guardianship in Neighborhood A ($r = -.16$, $p = .00$). Although informative in terms of the direction of the relationship, these results by themselves do not speak to the decision-making processes of the burglars. By contrast, the RTA task can shed light on the interpretation of the Easter Eggs and how they affect decisions. More than half of the participants spontaneously mentioned the Easter Eggs during their RTA task (see Table 5).

Table 5. Easter Egg quotes

"The waste containers here are also perfect. You can move them anywhere, climb on things, but also use them to hide behind." (Participant 149)

"They should never have placed this ladder here, of course. The ladder is an invitation to enter." (Participant 146)

"This window open here is just stupid. You wouldn't put a carport under an open window to a room, right? That's really foolish, foolish people. That's asking for a break-in."
(Participant 107)

"There are garbage bags outside here, either they're lazy or they haven't been here for a while. I see a parcel box laying outside, I definitely need to see what's inside or has been inside the box. Knowing that you can estimate the value of items the residents have in their house." (Participant 117)

The quotes in Table 5 illustrate the level of detail the RTA task is able to provide and provide a glimpse into the mechanisms that drive the decision-making process of offenders. Furthermore, they discussed how behavior can be discussed in more detail, capturing some of the reasons why participants pay attention to specific aspects in the environment.

Another interesting finding—building on the eye-tracking data and burglars’ interpretations—is that the use of VR-RTA enables access to information on which other aspects in a neighborhood burglars pay attention to and how they reflect upon those during the RTA task. This is information that would not have been uncovered by analyzing VR-system data (e.g., eye-tracking) only. It seems that many burglars based their assessment of the level of social control in a neighborhood on items lying outside houses (see Table 6). Many burglars have mentioned that in neighborhood A people tend to leave valuable items (e.g., kids’ toys, unlocked bikes or scooters, building material) lying outside their homes. Most interesting from a perspective of decision making on where to burgle is that they tell us during the RTA how this influences their decisions (see Table 6). They interpret that state of affairs first as a sign of occupancy. Moreover, they see it also as indicating that people do trust in a considerable level of social control in their area, which then acts as a warning for them as burglars. Also, toys out in the open are seen as signs of small children living here. Burglars then interpret that as a sign of raised guardianship, because people with small children as well as their neighbors tend to keep an eye open to watch over the kids.

Table 6. Quotes complementing eye-tracking data indicating what helps burglars estimate levels of social control.

Items in a neighborhood determining social control

“People feel safe in this neighborhood. They’re cool with leaving stuff outside. Not too many cars around. It’s nice for them, feeling safe and all. For me, it is convenient. People just leave things lying around in this neighborhood. It’s the kind of place where they’re comfortable, so you can tell from the vibe and all the stuff left outside. So, they will leave everything open, you know?” (Participant 189)

“Another thing to keep in mind when there are a lot of kids around is that people with children tend to keep an eye on others with kids and their children. So, in this neighborhood there are more eyes on the lookout.” (Participant 183)

“Over here, the area with construction materials – people don’t just leave that lying around casually; it’s quite noticeable. Either someone is keeping an eye on it, or if not, you could easily get in just with what’s here. Look, these folks have bought expensive stuff, so yeah, it’s almost inviting me to go inside. They probably aren’t paying as much attention to putting things away and locking doors when they’ve got pricey items.” (Participant 200)

“Young people in this neighborhood, you see toys and stuff scattered everywhere – scooters, it’s a residential area. Not an attractive neighborhood, I must say. You can be seen from everywhere. People with kids keep an eye out for suspicious characters in the street, so if you’re walking around here, you’re easily noticed. It’s quieter too, people are indoors earlier, so there are more folks at home. (...) Those who hire people for painting jobs leave their ladders lying around because everything is so safe here. So, if something fishy is going on, people are attentive.” (Participant 197)

The comments in Tables 5 and 6 provide in-depth information about the reasoning behind their gaze directions, which would otherwise have remained undiscovered. The burglars seized on multiple items in the neighborhood to determine attractiveness and gauge levels of social control, items that could be of importance in preventing burglary in real life.

Combining VR-RTA with interview questions

Results discussed in the previous sections suggest that the VR-RTA approach helps elicit information (i.e., providing information otherwise left undisclosed). In this section, we explore whether VR-RTA also enhances people's motivation to disclose information in addition to points raised in the interview (i.e., providing more in-depth information). In principle, similar to the analysis of CCTV footage, VR screen recordings can be replayed, paused, and slowed down an infinite number of times and provide a frame-by-frame account of the events as they unfold (Philpot et al. 2019). As a consequence, they allow for dissecting a decision-making sequence and for asking clarification questions in real time. However, unlike analyzing CCTV footage where researchers do not have access to those features in the footage, VR-RTA enables researchers to link observations to other relevant factors pertaining to individuals, including psychological states, motivations, dispositions, and background characteristics (van Gelder 2023).

In general, participants tended to give short, condensed answers during the semi-structured interview when asked directly about what attracted or deterred them in the virtual neighborhoods. By contrast, during the RTA, information tended to be more detailed, elucidating different steps in the decision-making process without necessarily requiring prompts. The VR-RTA findings also highlight the importance of environmental factors affecting decisions that are difficult to capture in traditional interviews that rely on memory.

By way of example, consider some of the differences between responses to the interview question "What attracts you in a neighborhood?" and the information elicited by the RTA task. In the interview, participant 50 responded: *"I only look at cameras,"* but the RTA from the same participant uncovered various details: *"There are several things I pay attention to. Windows that are open, gardens. It is difficult here because I can't look over the fences. I would do that to check if the residents were home. This neighborhood is difficult because everyone can monitor each other, you are in the line of sight from all the houses. You have to stay in the dark and avoid the lights. A lot to pay attention to."* Participant 72 responded in the interview: *"When I receive Tips or it's holiday time."* But in the RTA, the same participant responded: *"This dark alley is ideal for scouting, very attractive—easy hiding, limited police access, multiple escape routes. Good targets here. If there would be dogs and a lot of lights, those are a no-go"* The interview answer of Participant 94 was equally short: *"Wealth."* However, the RTA from the same participant revealed valuable insights: *"This neighborhood offers abundant hiding places, making it attractive for cover. Limited social activity means fewer people outside, with families as you can see from the toys lying around. Easy balcony access for break-ins, a small, long alley provides good cover. The quiet, boring atmosphere stands out; I prefer busy neighborhoods like in Amsterdam for blending in."*

In a similar vein, interview responses regarding deterrent factors in neighborhoods also tended to be brief. Conversely, during their RTA participants emphasized the importance of several facets determining surveillance, social control, and deterrence. Contrasted with the interview answer “*Nothing deters me*” from Participant 53, the RTA from the same participant revealed a more nuanced attitude: “*In a community where people watch and monitor each other, good neighbors are vigilant about suspicious activities, you have to be careful about the residents.*” Compared to the interview answer “*Nothing deters me, couldn’t care less,*” Participant 70 revealed different insights during the RTA: “*Vigilance and surveillance are problematic in this neighborhood. Close-knit neighbors oversee everything, making breaking in too risky. No, I wouldn’t break into this neighborhood, having too many eyes is dangerous.*” These examples demonstrate the importance of community dynamics in burglars’ decision-making processes, something which was not fully captured in traditional interviews. Similarly, the interview answer “*Nothing deters me, I can avoid everything*” from Participant 129 was nowhere near as insightful as the RTA from the same participant: “*Avoiding being visible is crucial. This neighborhood’s lack of attractiveness stems from houses being too close to each other, which enables residents to monitor each other closely.*” In the same way, the interview answer “*Nothing deters me, if it is worth it there is no fear*” from Participant 182), was complemented by insightful details during the RTA: “*Child-friendly neighborhoods pose risks. Vigilant parents, more people at home, and neighbors actively watching out make it too risky.*”

Taken together, the examples above demonstrate how the use of VR-RTA can provide valuable and detailed insights into the decision-making process of offenders. By allowing burglars to reflect on their actions using the RTA method, we tap into their cognitive process, making it easier for them to talk about their methods, and resulting in rich and detailed information. Figure 7 provides an overview of how the RTA task serves as the missing link to understanding criminal decision-making from the offender perspective.

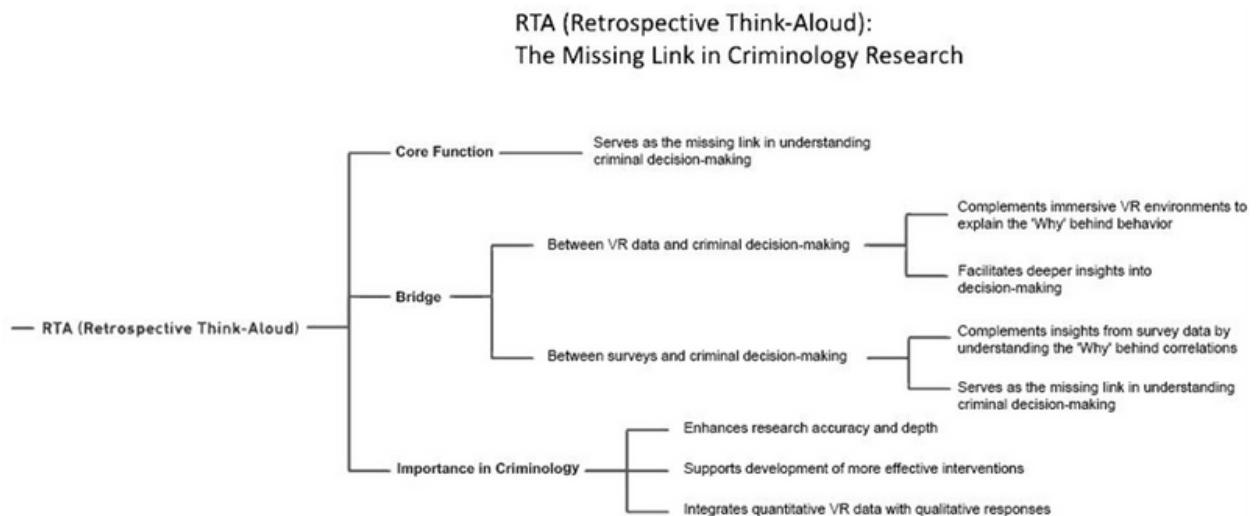


Figure 7. Overview of the RTA-method as the missing link in understanding decision-making and its importance in criminology research

Discussion

We introduced a novel method, VR-RTA, that integrates retrospective think-aloud protocols with virtual reality methodology to gain insight into offender decision-making processes. We illustrated the method among a sample of incarcerated burglars, who reflected on their actions from a first-person perspective after they had scouted a virtual neighborhood for opportunities to commit a burglary. Allowing burglars to reflect on their actions whilst observing their own experience makes it easier for them to elaborate on their choices and to verbalize cognitive processes while on the job. That is, the method allows for capturing “in-the-moment” considerations underlying decision-making during the crime commission process. The results demonstrate how the application of VR-RTA to the study of criminal behavior can provide relevant insights into the decision-making processes of offenders.

The VR-RTA method overcomes several challenges facing conventional approaches to studying criminal decision-making. First, by letting participants think out loud, it assists in the verbalization of decision-making processes that have become automated, and are therefore not subject to deliberate retrieval through interviews or surveys. Second, as the recording is played back immediately after the action, the time lag between the (virtual) crime and the subsequent data collection is reduced from what is commonly a period of months or even years to a mere few minutes, thus overcoming problems related to retrospection and memory degradation. Third, this approach also adds to virtual reality research and observational research as it allows for discussing specific participant behavior with them, rather than having to rely solely on behavioral or observational data. Lastly, VR-RTA turned out to be an efficient way to build rapport between the researchers and the participants.

The first research question regarded the extent to which the results in this study align with established findings from prior research into the cues that guide the decision-making of burglars. The four different categories of burglary cues (i.e., layout, security, occupancy, affluence) that had been established in earlier research also emerged in this study. In line with prior work, layout cues were most frequently mentioned with an emphasis on surveillance. Other frequently mentioned subcategories, such as social control, target hardening, and presence of residents, also aligned with prior work (Nee et al. 2019; Peeters 2013). These findings can be interpreted as supporting the validity of the VR-RTA method. In answer to the second research question, the VR-RTA method was shown to supplement interview and VR data in meaningful ways. First, the RTA task contributed to understanding the “why” behind gaze direction and spatial data. Second, the VR-RTA method elicited more detailed information, which may otherwise have been left undisclosed.

Limitations and Future Directions

We consider this study to be only the first step in the development of a novel and exciting approach to study criminal and antisocial behavior, having only scratched the surface in terms of exploring the possibilities of combining VR technology with think-aloud protocols. Inevitably, this study was also characterized by limitations that merit discussion. First, the were not able to quantify the extent to which our method enhanced participants' motivation to disclose information. Future studies should seek to quantify the degree the incremental value of the VR-RTA method, possibly by adding a control condition consisting of an interview but without a think aloud protocol. Secondly, it is important to acknowledge that committing a “virtual burglary” is not the same as committing an actual burglary. Despite the realism of the virtual neighborhoods in this study, and one of them being a virtual copy of an existing neighborhood, and despite prior research showing that burglars in virtual environments operate in similar ways as in real life (Nee et al. 2015, 2019; van Sintemaartensdijk et al. 2022), general transferability of (criminal) behavior in VR to behavior in the real world is not a given. Finally, participants did not physically walk through the virtual environment, but used a game controller to navigate it instead. Future studies could consider using treadmills so that participants actually engage in walking to render the experience even more realistic and immersive.

Conclusion

The VR-RTA method has strong potential as a tool to complement existing data collection approaches. It can help unveil cognitive process underlying criminal decision-making and allows researchers to access motivations behind specific behaviors that would otherwise remain undisclosed. In this way, the method provides a first-hand offender perspective and allows for examining crime in action. The present study serves as a first illustration of how think-aloud protocols can be integrated with VR in research designs. It is our hope that VR-RTA will be implemented across the field and become an important instrument in the criminological toolkit.

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Footnotes

1. The Virtual Burglary Project (VBP) is a collaboration between researchers from the Max Planck Institute for the Study of Crime, Security and Law (MPI) and the University of Portsmouth. In this research program, incarcerated burglars, and sometimes other groups of participants, are instructed to scope virtual neighborhoods for opportunities to burgle, select a target to burglarize, and/or to commit a burglary in VR (e.g., Nee et al., 2015, 2019; Van Gelder et al., 2017; Van Sintemaartensdijk et al., 2020, 2022). [↵](#)
2. To reduce participant burden, the scouting process was recorded in only one of the two neighborhoods. [↵](#)